

Python and C Programming

Task 02: Decision Making & Logic Building

Objective

The purpose of this task is to strengthen your understanding of variables, operators, user input, conditional statements, and logical thinking.

This task should be completed in both Python and C Programming.

Part A: Even or Odd Checker

Create a program that:

- Takes a number as input
- Checks whether it is Even or Odd
- Displays the result

Example

Input:

15

Output:

15 is an Odd Number

Part B: Largest of Three Numbers

Create a program that:

- Takes three numbers as input
- Finds the largest number
- Displays the result

Example

Input:

10
25
18

Output:

Largest Number = 25

Part C: Student Grade Calculator

Create a program that:

- Accepts marks from the user
- Displays grade according to the following criteria

Marks	Grade
90-100	A
80-89	B
70-79	C
60-69	D
Below 60	F

Example

Input:

85

Output:

Grade: B

Part D: Simple Login Validation

Create a program that:

- Stores a predefined username and password
- Takes input from the user

- Validates the credentials

Example

Username:

admin

Password:

password123

Output:

Login Successful

Otherwise:

Invalid Credentials

Part E: Mini Cyber Security Challenge

Create a Password Strength Checker.

The program should check:

- Password length should be at least 8 characters
- Contains at least one number
- Contains at least one uppercase letter

Display:

- Weak Password
- Moderate Password
- Strong Password

Example:

Password: Cyber123

Output:
Strong Password

Documentation

Create a folder named:

Programming_Task_02_YourName

Inside the folder include:

Python Files

- even_odd.py
- largest_number.py
- grade_calculator.py
- login_validation.py
- password_checker.py

C Files

- even_odd.c
- largest_number.c
- grade_calculator.c
- login_validation.c
- password_checker.c

Also include:

- Screenshots of execution
- README.md
- Brief explanation of each program

Upload everything to GitHub.

Submission

The goal of this task is to improve problem-solving and programming logic.

Submit your updated GitHub repository link through the Sunday Submission Form before the weekly deadline.

Continue maintaining all internship tasks in the same repository using separate folders.

Topics to Study Before the Next Session

Please spend some time learning:

Python

- if, elif, else
- Comparison Operators
- Logical Operators
- String Functions
- User Input Handling

C Programming

- Decision Making Statements
- if-else
- Nested if
- Relational Operators
- Logical Operators
- Basic String Handling

Common Concepts

- Flowcharts
- Problem Solving
- Algorithm Basics
- Input Validation

Recommended Study Time:

30–45 minutes per day.

Remember: Strong programmers are not those who memorize syntax; they are those who can break a problem into smaller steps and solve it logically.